

Intelligent Surveillance and Identification for Secure Campus

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Problem Statement

Current Challenges in Campus Security

- > Manual surveillance is labor-intensive and prone to human error
- > Unauthorized access and identity misrepresentation risks
- > Lack of automated verification mechanisms
- > Difficulty in monitoring large student populations across expansive campuses

Traditional methods cannot scale effectively with growing student populations and complex campus infrastructures

Project Objectives

Primary Goals

- > Design AI-powered surveillance system
- > Automate student identification using facial recognition
- > Enforce uniform compliance through color detection
- > Real-time alert generation for violations

System Features

- > Web-based admin dashboard
- > Centralized database integration
- > Minimize human intervention
- > Detailed logging

System Architecture

1

Video Acquisition

CCTV cameras capture live streams, frames extracted and preprocessed



2

Facial Recognition

DeepFace model generates 128-dimensional embeddings for identity verification



3

Uniform Detection

Custom CNN detects shirts, K-Means clustering extracts dominant color



4

Cross-Verification

Compare detected department with database records



5

Alert Generation

Real-time notifications for violations displayed on dashboard



Cross-Verification Process

Verification Conditions

Detected Face Department	Detected Uniform Department	System Response
Unknown	Unknown	 Outsider Detected
Unknown	Known	 Potential Security Threat
Known	Unknown	 Not in University Uniform
Department A	Department B	 Wrong Department Uniform
Match	Match	 Verified Successfully

Applications & Features

Current Features

- > Campus entry monitoring
- > Uniform compliance enforcement
- > Departmental classification
- > Real-time violation alerts
- > Administrative logging

Potential Extensions

- > Automated attendance system
- > Behavioral analytics
- > Mobile app notifications
- > Access control integration
- > Historical trend analysis

Limitations & Challenges

Technical Limitations

- > **Lighting Variability**: Performance affected by poor lighting or harsh shadows
- > **Camera Placement**: Extreme angles or distances reduce accuracy
- > **Motion Blur**: Fast movement causes detection challenges
- > **Video Quality**: Low resolution affects facial and color recognition

Operational Challenges

- > **False Positives**: Similar uniform colors may cause misclassification
- > **Hardware Dependency**: Needs quality CCTV infrastructure and System with higher specification (primarily GPU)
- > **Occlusions**: Partially visible faces or uniforms reduce accuracy

Conclusion

Project Achievements

- > Successfully developed AI-driven campus surveillance system
- > Integrated DeepFace for accurate facial recognition
- > Implemented custom CNN for shirt detection
- > Created real-time cross-verification mechanism
- > Built interactive dashboard for monitoring
- > Demonstrated moderate accuracy in testing and validation

Key Impact

This system transforms traditional manual surveillance into an intelligent, automated solution that enhances campus security, enforces discipline, and improves operational efficiency through AI and computer vision technologies.

Thank You!